

Tracking Earth's Changes with AlphaEarth Foundations

[Git Repo Here](#)

Alpha Team

Oct 12, 2025



Alpha Team



Dr. Brendon Hall
*Sr. Manager in AI for Energy
& Utilities, Deloitte*

Our Mentor

integrating physics
based models, ML and
DS to create
transformative software
tools for the oil and gas
industry.

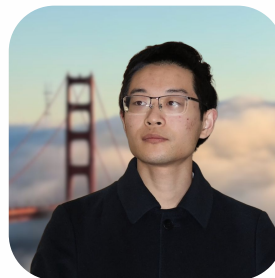


Xihan Yao
1st year GRG PhD student

Our Developer

Remote Sensing/GIS

Human-Environment
Interactions



Koutian Wu
2nd year GEO PhD student

Our Developer

Develop the Noah-MP
land surface model

Understand the land
surface by integrating
physics and AI



Rushi Bhatt
*Senior EVS (GRG)
Undergraduate*

Our Developer

Write Code.
Test Code.
Push Code.
...
Save The Planet.
(some steps may be missing).
GeoAI, ML,
Sediment/Runoff Routing

Song: Changes

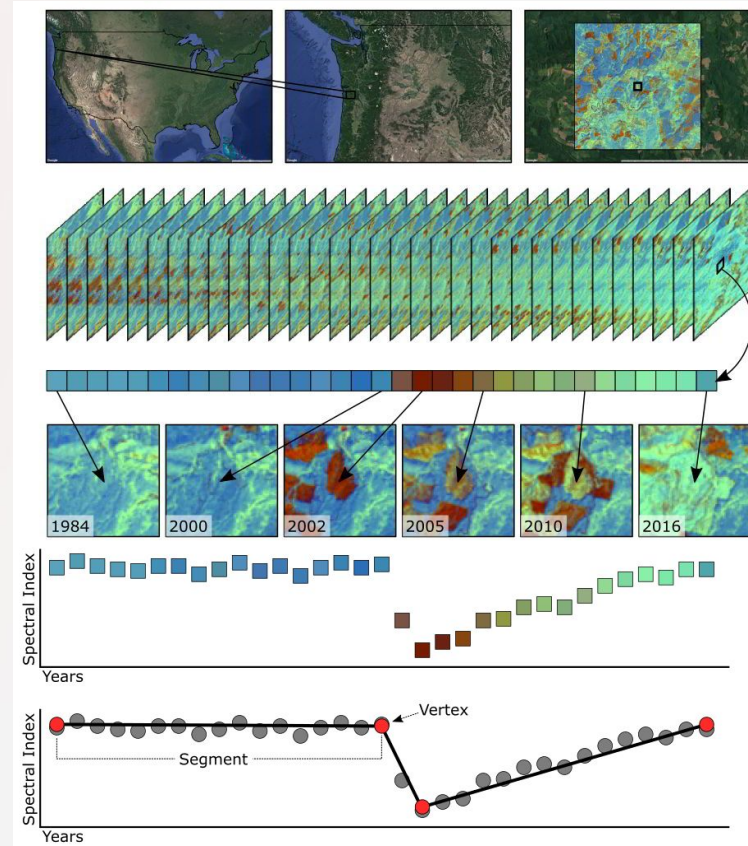


Click the image to play

An accessible tool to Our Changing World

It is not about where changes happened. But also about the magnitude, the duration, etc

Our goal: a user friendly tool to identify the changes, **When did it happened?**
How drastic was the change? How long did it last?

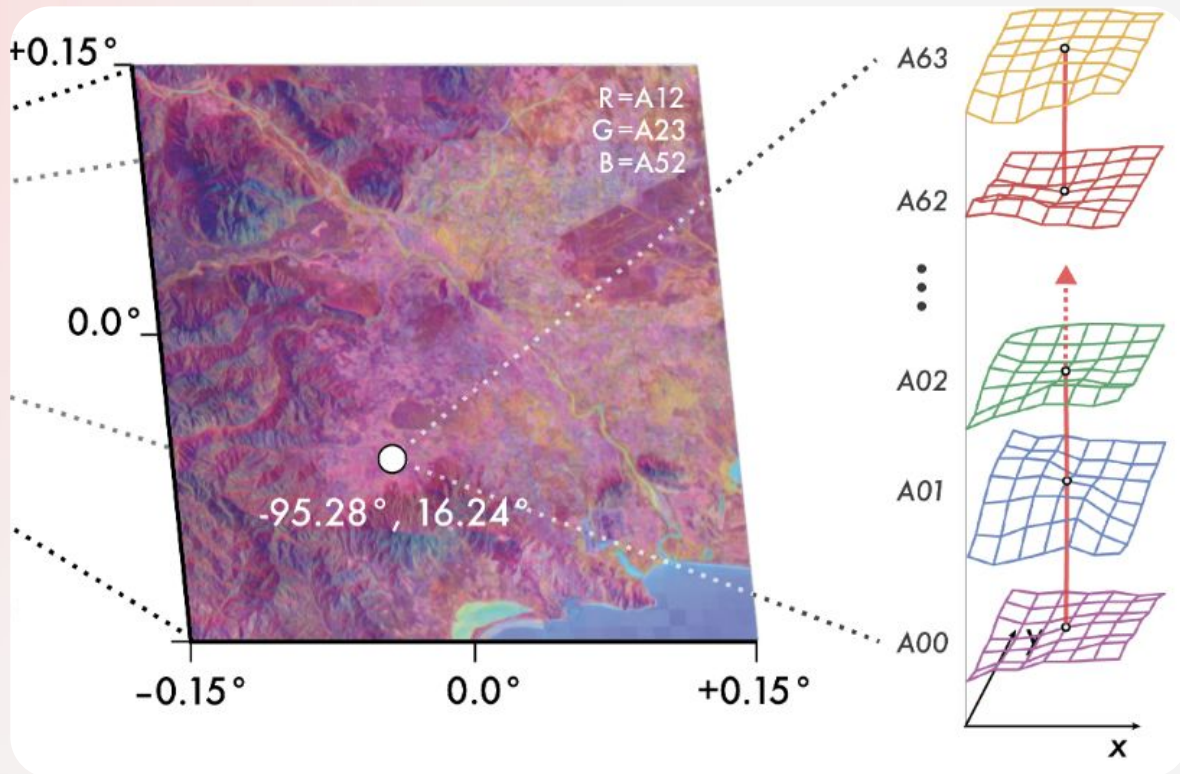


AlphaEarth Foundations

Global embedding model that captures Earth's surface at 64 layers, 10m resolution

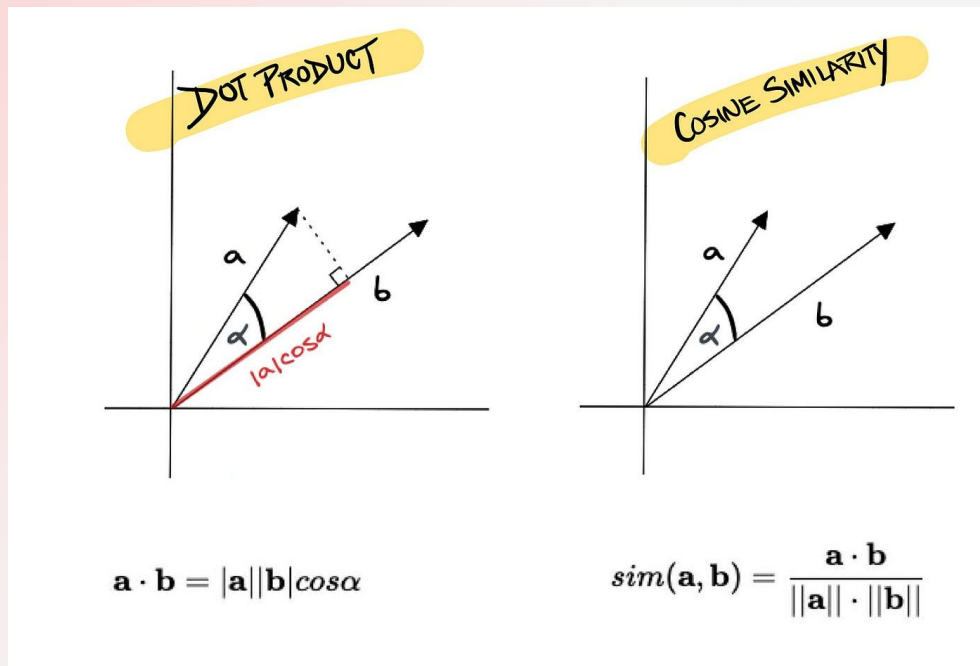
Trained on **3+ billion** observations from satellites, radar, LiDAR, climate data, geotagged information, etc

Creates "universal features" that work across different mapping tasks

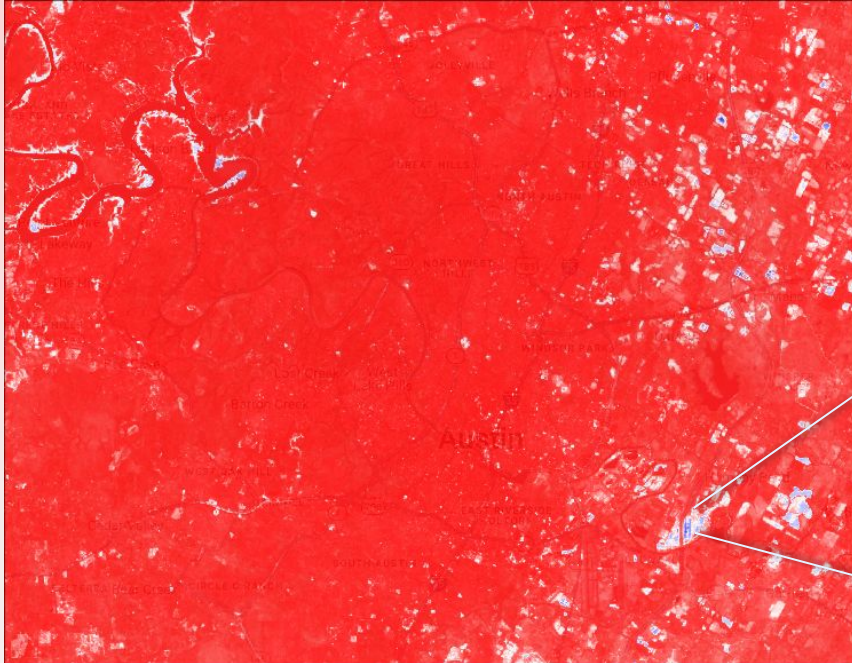


Brown et al., 2025

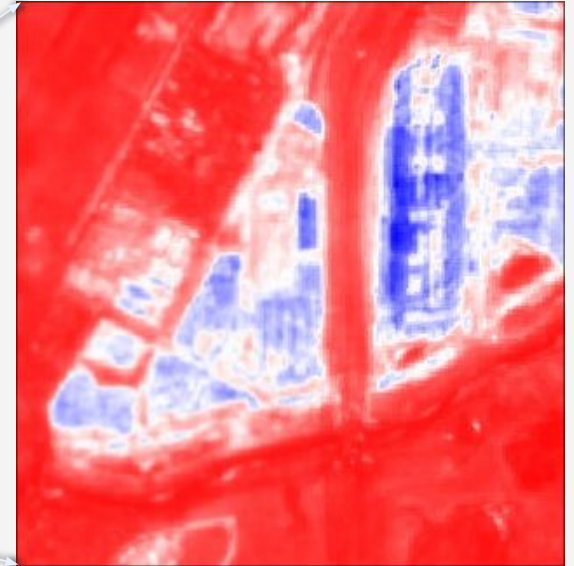
Cosine similarity concept



Cosine similarity for Austin Area



Greater Austin Area (2017 → 2025)
Cosine Similarity of 64-Dim
(red = no change, white = some change, blue = lot's of change!)



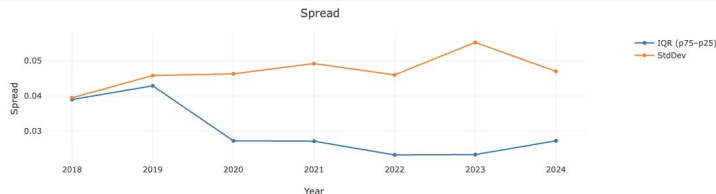
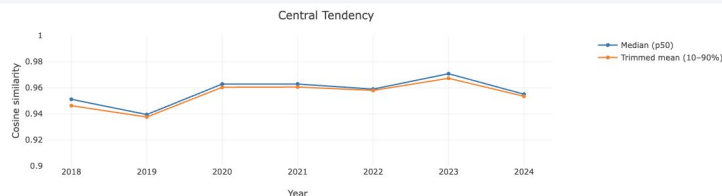
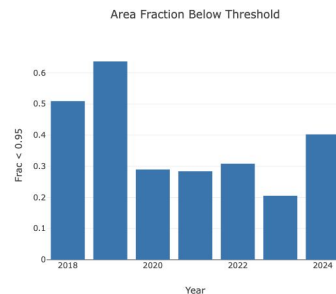
Try your own! See how Austin embedding changed



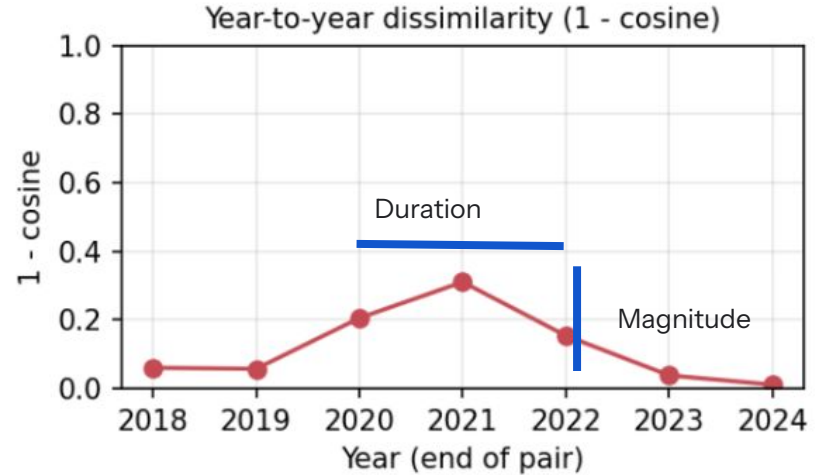
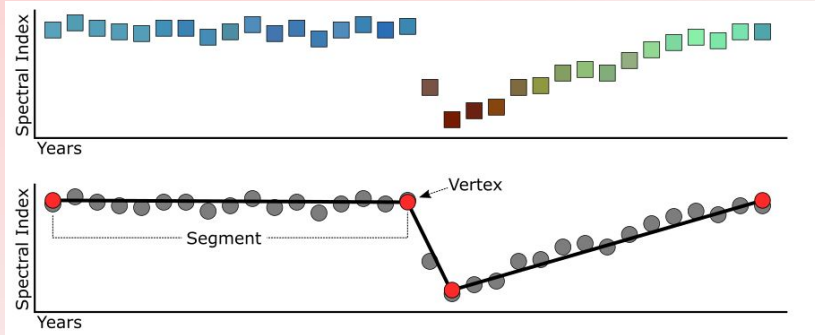
Scan QR code or type into your browser:

<https://ktwu01.github.io/AlphaEarthHack>

AlphaEarth Austin Change Analysis (2017-2024)



Alpha Earth Change Detection



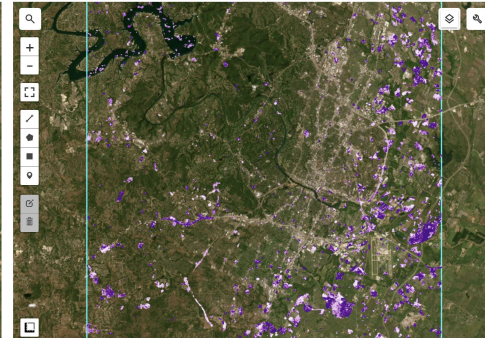
LandTrendr Alpha Earth Comparison

Year of Detection

Magnitude

Duration

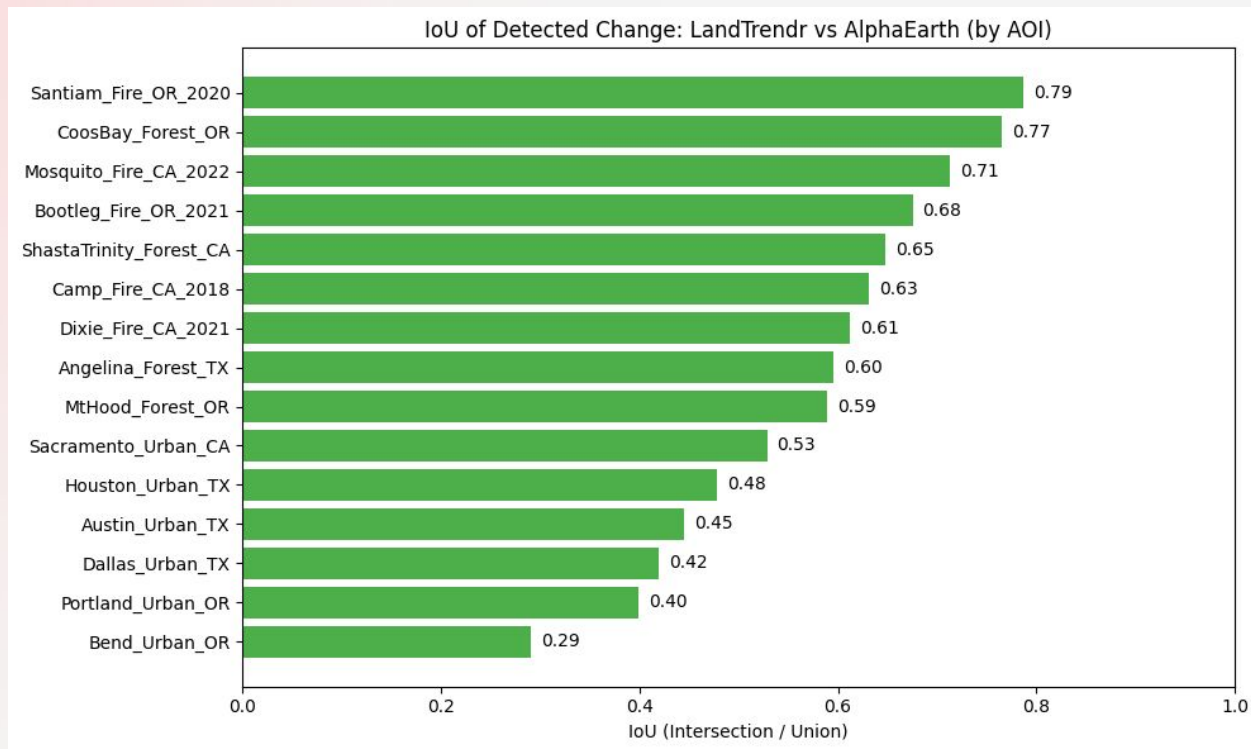
landTrendr



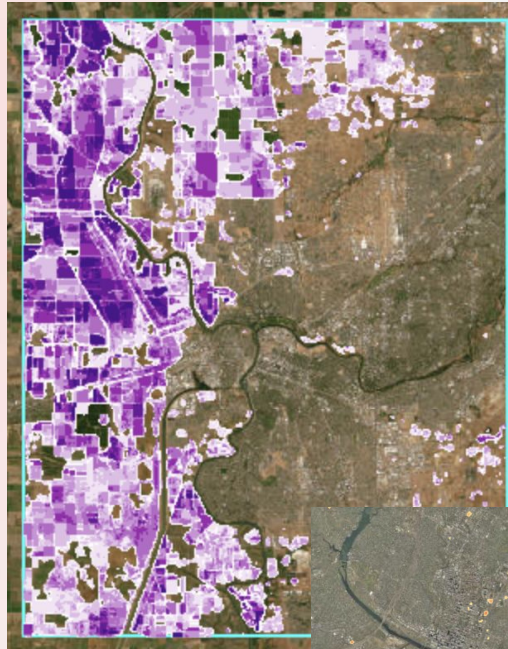
AlphaEarth



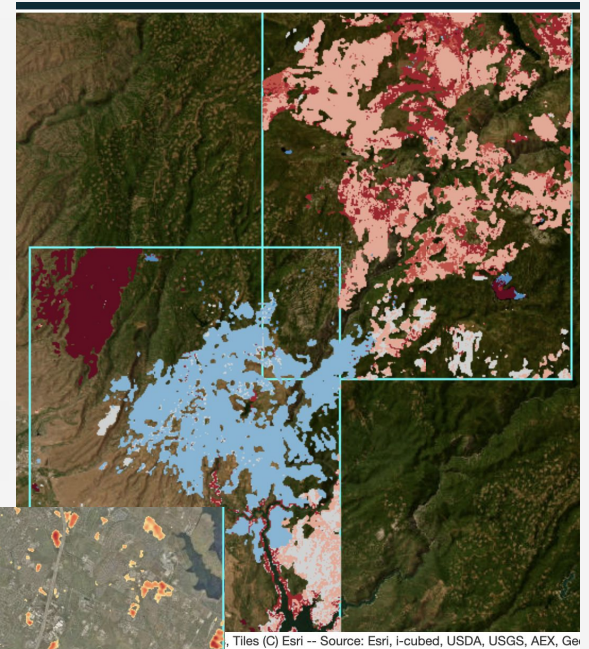
LandTrendr vs Alpha Earth Comparison



Urban, Fire and Logging



Duration,
Sacramento,
CA



Year of Detection, Fires,
CA

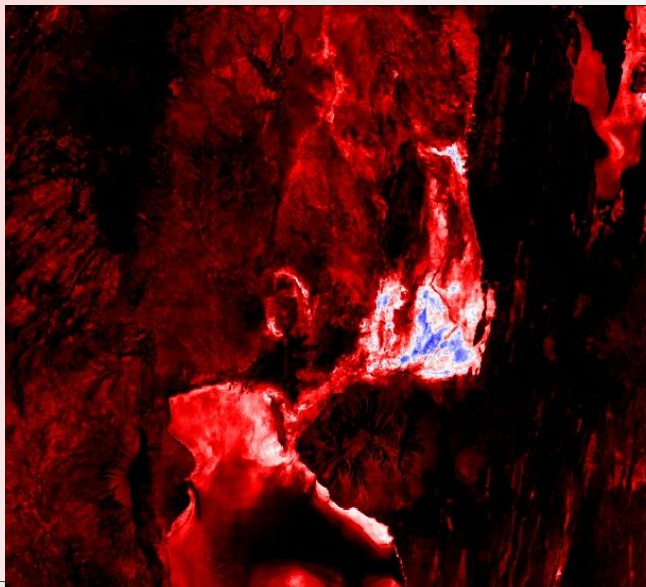


Magnitude,
Austin, TX

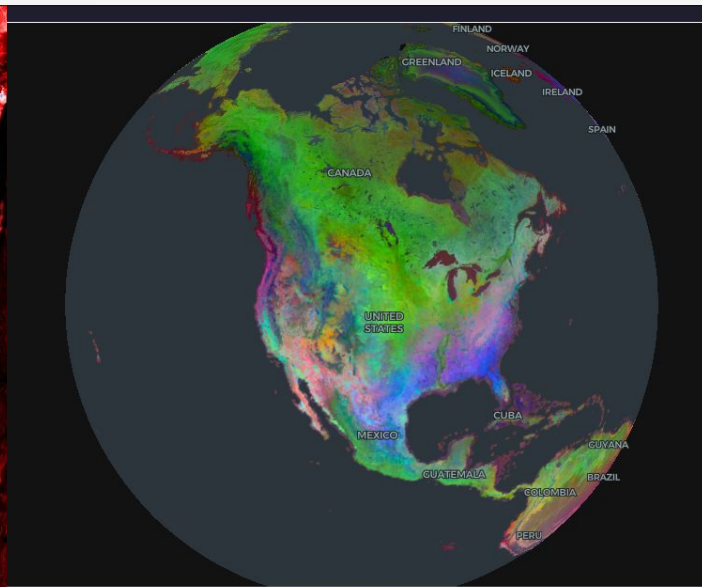
Story Time



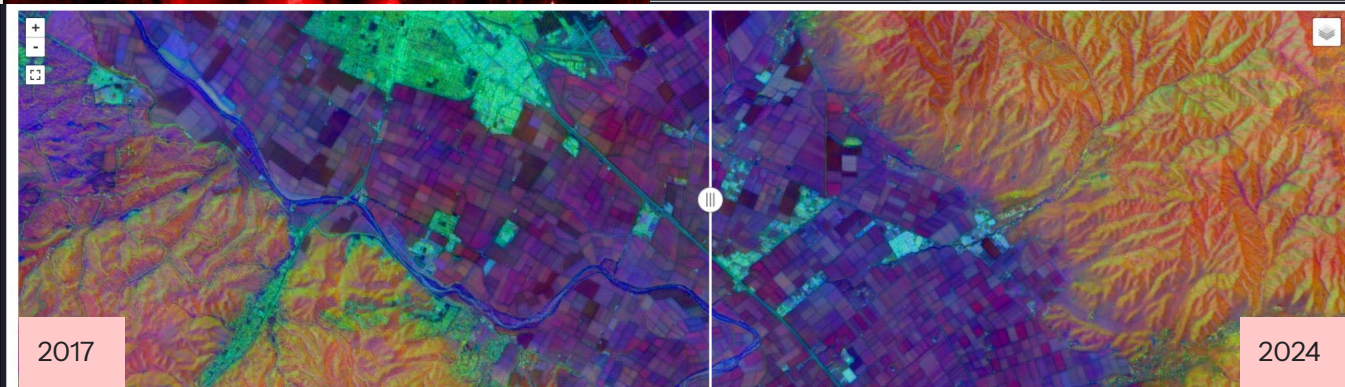
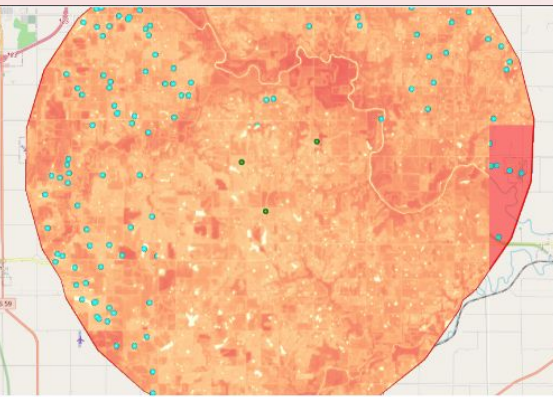
Cosine Similarity



False-Color Imagery



Object Search within Locality



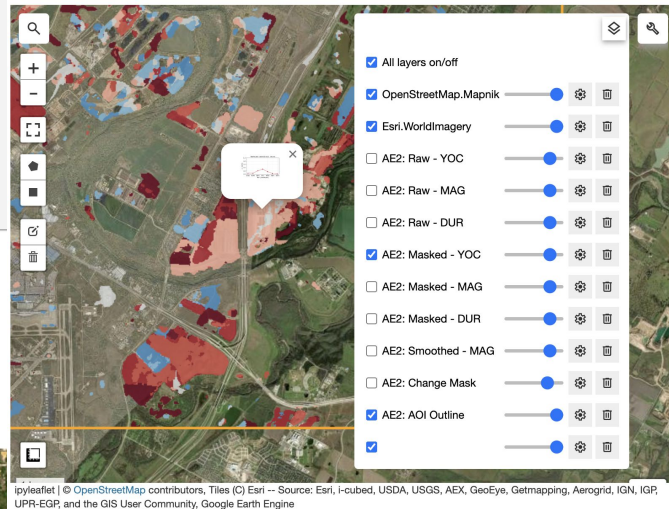
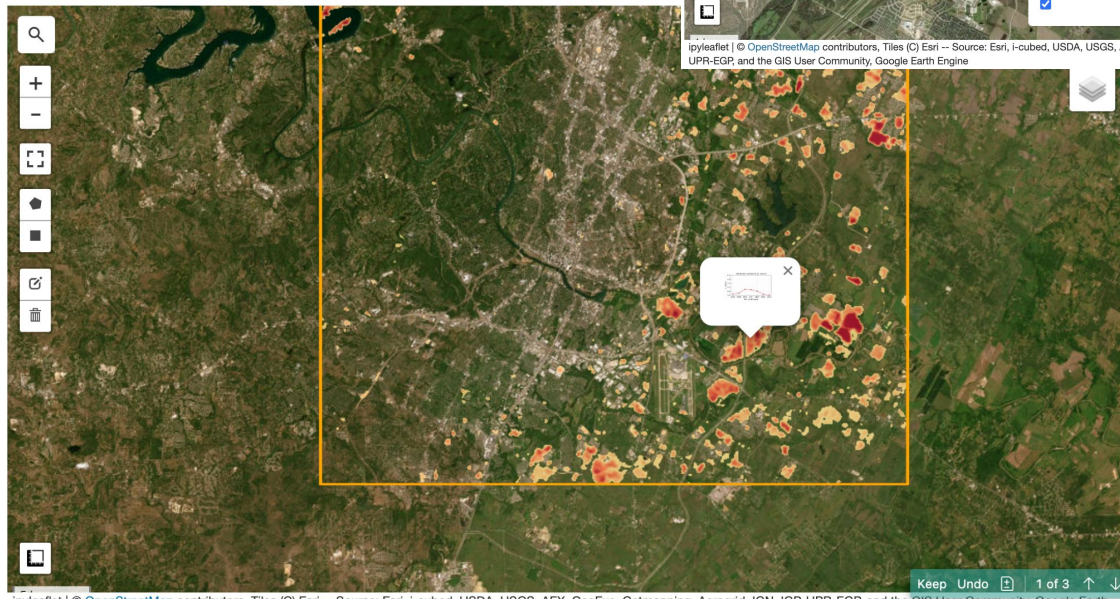
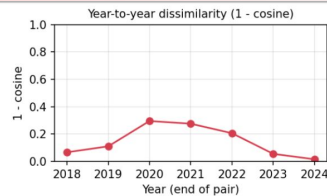
see?

—An Interactive tool

Visualization

Inspection

Analysis-ready data
preparation



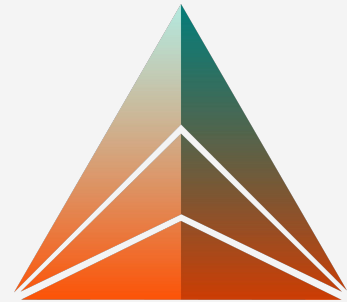
Open Source Software



NumPy



plotly



Plan: make an accessible toolkit to facilitate geoscience community

The screenshot shows the GitHub repository for AlphaEarthHack. The repository is public and has 1 star and 0 forks. The main branch is selected. The file list includes backup, images, .gitignore, AlphaEarth_Interactive_Mapping.ipynb, AlphaEarth_LandTrendr_ChangeComparison..., AlphaEarth_Story.ipynb, Criteria.md, LICENSE, LandTrendr_AlphaEarth.ipynb, README.md, alpha_tutorial.ipynb, and austin_change.ipynb. The README is selected, showing the title 'AlphaEarthHack - UT Austin Geoscience Hackathon' and the BSD-3-Clause license. The right sidebar shows the About section with the team name 'AlphaEarthHack' and a link to the repository. It also lists 1 star, 0 watching, and 0 forks. The Releases section shows no releases published. The Packages section shows no packages published. The Contributors section shows 5 contributors. The Deployments section shows 5 deployments, including github-pages 12 minutes ago.

AlphaEarthHack Public

Pin Watch 0 Fork 0 Star 1

main 3 Branches 0 Tags

Go to file Add file Code

God-Is-A-Bird Update README.md 70557a1 · now 37 Commits

backup	Add training data and statistics for various assets	3 hours ago
images	Add files via upload	21 minutes ago
.gitignore	Added Sush's test script @GEE-test.ipynb	2 days ago
AlphaEarth_Interactive_Mapping.ipynb	Implement code changes to enhance functionality and im...	1 hour ago
AlphaEarth_LandTrendr_ChangeComparison...	Enhance AlphaEarth EDA notebook and add LandTrendr c...	12 hours ago
AlphaEarth_Story.ipynb	Add files via upload	2 hours ago
Criteria.md	feat(readme):Add new Jupyter notebook for AlphaEarth e...	46 minutes ago
LICENSE	Initial commit	2 days ago
LandTrendr_AlphaEarth.ipynb	Add training data and statistics for various assets	3 hours ago
README.md	Update README.md	now
alpha_tutorial.ipynb	Update change detection workflow: server-side EE ops, d...	14 hours ago
austin_change.ipynb	feat(readme):Add new Jupyter notebook for AlphaEarth e...	46 minutes ago

README BSD-3-Clause license

AlphaEarthHack - UT Austin Geoscience Hackathon

About

UT Austin Geoscience Hackathon:
'AlphaEarthHack' team

ktwu01.github.io/AlphaEarthHack/

Readme
BSD-3-Clause license
Activity
1 star
0 watching
0 forks

Releases

No releases published
[Create a new release](#)

Packages

No packages published
[Publish your first package](#)

Contributors 5

Deployments 5

github-pages 12 minutes ago

Thank you

Questions?

